EVEREST ENGINEERING COLLEGE

(Affiliated to Pokhara University)

Sanepa-2, Lalitpur



ASSIGNMENT

OF

SOFTWARE ENGINEERING

Submitted by

MAMATA YADAV

PRAKRITI GIRI

Submitted to

ER. BIRODH RIJAL

DEPARTMENT OF COMPUTER ENGINEERING

Assignment 1

The exam control office of Pokhara university aim to digitize its services, making all required task accessible online. The standard operating procedures for each task are already well-defined. While the project has a reasonable timeline for completion, it must be executed within a limited budget.

Which software process model would be more suitable for developing this software? Analyze and discuss the advantage of the chosen process model.

- According to the question scenario, waterfall model will be suitable for developing this software. As we know that, the waterfall model is useful in situations where the project requirements are well-defined and the project goals are clear. The waterfall model is used in situations where there is a need for a highly structured and systematic approach to software development. It can be effective in ensuring that large, complex projects are completed on time and within budget, with a high level of quality and customer satisfaction.

  In the context of our project advantages of using waterfall model are:

  - Suitable for projects when the requirements are clear and stable.
  - It is easy to understand and implement because each phase is processed one at a time.
  - Waterfall Model reinforces good habits like define-before-design and design-before-code.
  - It is very simple and easy to understand.

  Disadvantages of using waterfall model are:

  - This model recommends that a new phase can start only after the completion of the previous phase. To increase efficiency and reduce cost, phases may overlap.
  - It is a rigid and linear approach to software development, which means that it is not well-suited for projects with changing or uncertain requirements.

- There is no feedback in this process model. If the delivered system doesn't meet expectations, it results in wasted time and effort.

  Hence, Waterfall model might be suitable for a digitize the services of exam controller system with well-define requirements.

Assignment 2

• A Flight Control Software System is a specialized, safety-critical software designed to manage and regulate an aircraft's flight control

surfaces, such as ailerons, rudders, elevators, and flaps. Its primary purpose is to ensure stability, maneuverability, and safety during all

phases of flight, including takeoff, cruising, and landing.

• Key Functions:

1. Automatic Flight Control: Supports autopilot operations to maintain altitude, heading, and speed.

2. Manual Flight Assistance: Enhances pilot control by interpreting input from the cockpit controls and translating them into precise movements of the

flight surfaces.

3. Flight Envelope Protection: Ensures the aircraft operates within safe limits by preventing dangerous maneuvers, such as stalling or exceeding

structural stress thresholds.

4. Sensor Integration: Gathers data from sensors (e.g., airspeed, altitude, and gyroscopic data) to dynamically adjust control actions.

5. Redundancy and Fault Tolerance: Incorporates multiple levels of redundancy to continue functioning in case of component or system failure.

• Importance:

• Flight control software systems are critical for modern fly-by-wire aircraft, where traditional mechanical and hydraulic controls are replaced or

augmented by electronic systems. Their performance and reliability are essential to ensure passenger safety and compliance with stringent aviation

standards like DO-178C (Software Considerations in Airborne Systems and Equipment Certification)

Which software process model would be most suitable for developing this software? Analyze and discuss the strengths and weaknesses of the chosen process model.

- According to the question scenario, spiral model will be suitable for developing the flight control system software. As this, model emphasis on risk management and iterative development. Spiral model can be an effective approach for developing safe, reliable and efficient flight control software. The critical nature of the system demands a robust risk management approach and the ability to adapt to changing requirements.

Strengths of the chosen process model are:

- The spiral model explicitly incorporates risk assessment and mitigation throughout the development process. It also helps to minimize costly errors and delays by identifying and addressing potentials risks early on.
- This model allows for changes in requirements or designs as the project progresses.
- This model also encourages regular customer feedback and involvement which ensure that the software meets the specific needs and expectations of the stakeholders, including pilots, air traffic controllers and regulatory authorities.

Weaknesses of the chosen process model are:

- This model can be more complex to manage because it requires a high level of expertise in risk management and iterative development.

- As this model includes multiple cycles of development and testing, which lead to higher development costs and delay in projects compared to other models.

  Hence, Spiral model might be suitable for developing the flight control system software with well-define requirements.

Assignment 3

Imagine developing a smart tourist guide system that provides end-to-end services for tourists. This will be a public application accessible to all users. The goal is to create engaging, value-added features while building a community and ensuring sustainability through revenue generation. However, the specific features that will attract and benefit users are yet to be identified.

Which software process model would be more suitable for developing this software? Analyze and discuss the strength and weakness of the chosen process model.

- For developing a smart tourist guide system with engaging, community-driven, and value-added features, where the specific features and user needs are yet to be identified, the Agile Model is the most suitable software process model. The ability to adapt to user feedback, focus on delivering value, and build a community around the system are crucial for the success of such a project. By carefully managing the project and addressing the potential challenges, the agile model can be an effective approach for developing a successful smart tourist guide system.

  Strengths of the chosen process model are:
  - It's able to adapt to changing requirements and feedback.
  - Delivers value to the user early and often.

- Fosters community and collaboration.

Weaknesses of the chosen process model are:

- Without proper management, the iterative nature can lead to scope creep.
- The changing nature of requirements can make it challenging to estimate costs and timelines accurately.
- Successful implementation of agile requires a highly skilled and motivated team.

Hence, Agile model might be suitable for developing a smart tourist guide system software.